

GEORGE BROWN POLYTECHNIC

Applied A.I. Solutions Development

FINANCIAL & OPERATIONAL BENCHMARKING OF U.S. HOSPITALS

Using CMS HCRIS Cost Report Data (2010-2024)

COMBINED PROJECT REPORT

Group data-visualization project focused on financial and operational benchmarking of U.S. hospitals using CMS HCRIS cost-report data.

PROJECT INFORMATION

Project Type	Group Project
Prepared by	Group 3
Institution	George Brown Polytechnic
Program	Applied A.I. Solutions Development
Project	MedInsight Analytics Inc.
Coverage	Case Study Set-up Advanced Analytics

GROUP MEMBERS

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PROJECT PROFILE

Industry	Healthcare - Hospital Financial Management and Analytics
Dataset	CMS HCRIS Hospital Cost Report Data
Period	2010-2024
Raw Records	120,238 hospital-year records
Hospitals	4,762
States / Territories	56
Variables	32
Advanced Methods	Regression, Pareto, k-means, TabPy z-score, distribution analysis, rolling averages

CASE STUDY SET-UP REPORT

A.1 Company Start-up

This report is built on a fictional start-up company named MedInsight Analytics Inc., a healthcare data-analytics company headquartered in Toronto, Ontario. MedInsight provides financial and operational benchmarking services to hospitals, health care systems, and consultants across the United States. Its core product is a subscription analytics platform that transforms publicly available Medicare cost-report filings into decision-ready dashboards and benchmark reports. It allows a hospital CFO or COO to compare their facility's financial health and resource utilization against true peer companies (same state, ownership type, bed size, and teaching status).

A.2 Industry / Sector Selection

Selected industry: Healthcare - hospital financial management and analytics. The healthcare sector was chosen because it has got rich data, it is heavily regulated, and economically significant. Moreover every Medicare-certified hospital in the United States is legally required to file a detailed annual cost report. It creates a consistent, publicly available dataset.

A.3 Main Data Source

The main data source is the Healthcare Cost Report Information System (HCRIS) - the CMS repository of Medicare cost reports filed annually by every certified hospital (Form CMS-2552-10). The working file used for this case study, `Final_dataset_hcris_hospital_cost_reports_2010_2024.xlsx`, is a cleaned, analysis-ready extract of HCRIS in XLSX format containing 120,238 hospital-year records for 4,762 hospitals across 56 states/territories, fiscal years 2010–2024, with 32 variables covering facility characteristics, revenues, expenses, margins, uncompensated care, and utilization.

Data source names and links

- CMS — Cost Reports (official HCRIS raw files, by fiscal year): <https://www.cms.gov/data-research/statistics-trends-and-reports/cost-reports>
- CMS — Hospital 2552-2010 form data (2010+ cost reports): <https://www.cms.gov/data-research/statistics-trends-and-reports/cost-reports/hospital-2552-2010-form>
- data.cms.gov — Hospital Provider Cost Report (curated annual dataset): <https://data.cms.gov/provider-compliance/cost-report/hospital-provider-cost-report>
- RAND Hospital Data (cleaned HCRIS in Excel/CSV — variable names match this dataset): <https://www.hospitaldatasets.org/>
- NBER — HCRIS Hospital Cost Reports mirror: <https://www.nber.org/research/data/hcris-hosp>

Dataset profile

Property	Value
File / format	Final_dataset_hcris_hospital_cost_reports_2010_2024.xlsx (Excel)
Records	120,238 raw rows (61,410 unique hospital-year filings after removing duplicate CY/FY rows)
Hospitals (prvdr_num)	4,762 unique Medicare provider numbers
Coverage	Fiscal years 2010–2024, 56 U.S. states & territories
Hospital types	Short-Term Acute Care (83,944), Critical Access (34,696), Children's (1,598)
Ownership (control)	Voluntary non-profit, proprietary, and governmental categories
Teaching status	32,462 teaching hospital-years; 87,776 non-teaching
Health systems	615 named systems (AHRQ Compendium linkage)
Variables	32 (identity, location, period, financial, utilization, staffing)

The source is authoritative (CMS), public and free, longitudinal (15 years), sufficiently large for meaningful analysis, and relational in nature (hospitals → annual reports → metrics), which supports normalization and database design. It is not a retail or e-commerce dataset.

A.4 Functional Areas of Focus

The case study focuses on two functional areas:

- Finance - net patient revenue, operating and other expenses, net income, operating margin, excess margin, uncompensated-care cost, DSH and medical-education funding.
- Operations - total beds, inpatient bed-days, inpatient discharges, occupancy, total employees, and residents/interns (staffing intensity).

These two areas are natural for cost-report data: the financial statement fields support profitability and sustainability analysis, while the utilization fields support capacity and productivity analysis, and combining them yields efficiency measures such as revenue per discharge, cost per bed-day, and employees per occupied bed.

A.5 Objectives of the Case Study

- 1 - Revenue-stream research: quantify the composition and growth of hospital revenue (net patient revenue vs. other and non-operating income) by year, state, ownership type, and teaching status, 2010–2024.
- 2 - Profitability monitoring: track operating margin and excess margin over time and flag structurally unprofitable hospitals.
- 3 - Uncompensated-care analysis: measure uncompensated-care cost (uc_cost) burden relative to net patient revenue and its relationship with funding across ownership types.
- 4 - Operational efficiency: analyze bed occupancy ($\text{ip_bed_days_total} \div (\text{beds_total} \times 365)$), discharges per bed, and employees per occupied bed to benchmark capacity utilization and staffing productivity.
- 5 - Peer benchmarking product: deliver percentile-based peer-group benchmarks that feed MedInsight’s client dashboards and annual benchmark reports.

A.6 Process & Data Models

A.6.1 Data Flow Diagram (Level 0)

The DFD below shows how MedInsight ingests the annual HCRIS extract from CMS/RAND, validates and stores raw filings, cleans and standardizes them into a hospital master store, computes financial and operational KPIs into a KPI warehouse, and serves benchmark reports and dashboards to hospital clients and analysts.

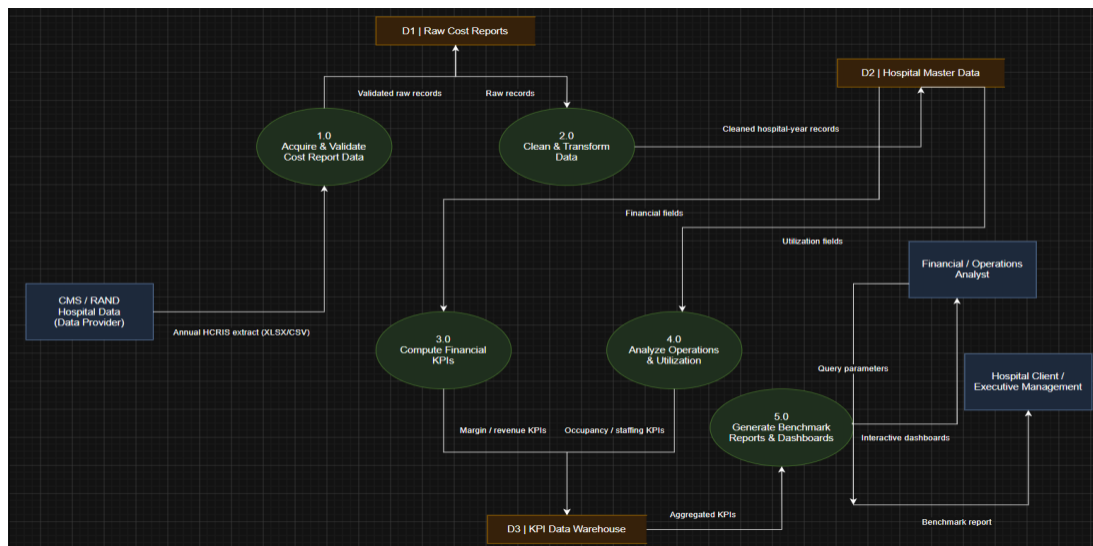


Figure 1. Level-0 DFD

A.6.2 Entity-Relationship Diagram (2NF)

The source file is a single flat table keyed by the composite key (prvdr_num, year). In the flat form, hospital-level attributes (name, address, city, state, coordinates, type) and the health-system name depend only on prvdr_num. The design below removes all partial dependencies.

- HEALTH_SYSTEM (health_sys_id) - one row per health system (615 systems).
- HOSPITAL (prvdr_num) - one row per facility; attributes fully dependent on prvdr_num alone.
- COST_REPORT (prvdr_num, year) - one row per annual filing after removing duplicate CY/FY reporting entries: reporting period, ownership control, and teaching status, which can change year to year.
- FINANCIAL_METRICS (prvdr_num, year) - revenue, expense, margin, uncompensated-care, and funding fields; every attribute depends on the full composite key.
- UTILIZATION_METRICS (prvdr_num, year) - beds, bed-days, discharges, employees, residents/interns.

*Note: the raw HCRIS extract contains two rows per hospital-year (one tagged reporting = 'CY', one 'FY'), which are otherwise identical. (prvdr_num, year) Therefore, only becomes a true primary key for COST_REPORT after filtering to a single reporting type — this is applied as a data-source-level filter before the model above holds.

Relationships: HEALTH_SYSTEM 1–M HOSPITAL; HOSPITAL 1–M COST_REPORT; COST_REPORT 1–1 FINANCIAL_METRICS and 1–1 UTILIZATION_METRICS.

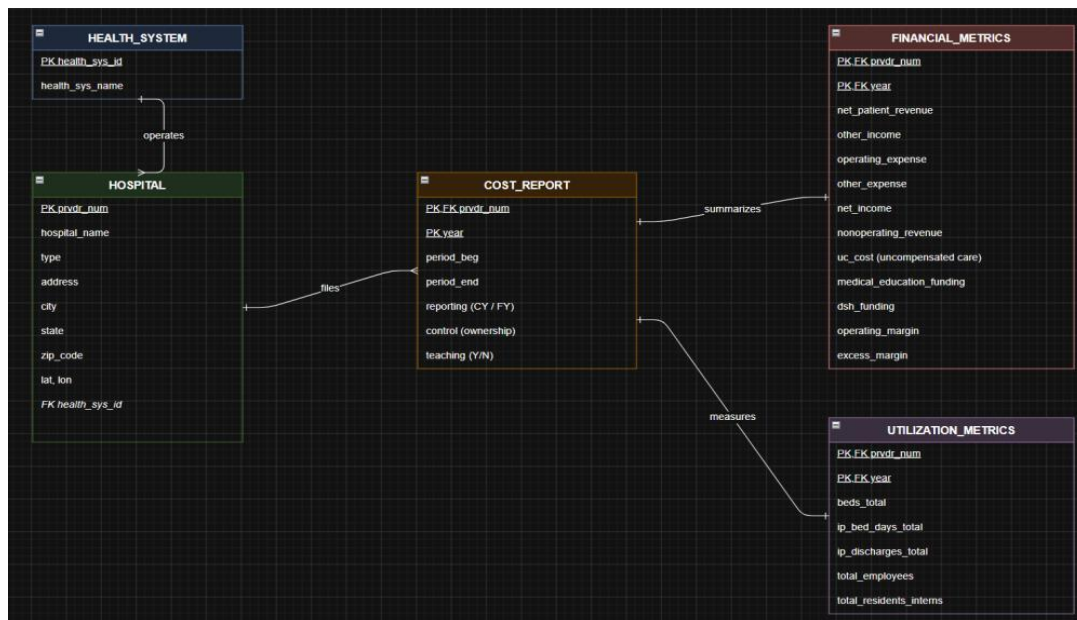


Figure 2. ERD in Second Normal Form

ADVANCED ANALYTICS

This exercise builds on MedInsight Analytics Inc.'s hospital benchmarking work, applying more advanced statistical and machine-learning methods in Tableau. Specifically, we set out to:

- See how hospital bed capacity relates to revenue, and whether that relationship changes depending on facility type
- Find which states carry the largest share of uncompensated-care costs
- Group hospitals into performance clusters based on occupancy and staffing levels
- Flag hospitals with unusual year-over-year expense growth using a Python-based anomaly detection model
- Look at how staffing intensity varies across hospital types
- Understand the sector's financial resilience over time — separating real structural trends from one-off yearly swings — and check whether revenue growth actually translates into better operating margins.

To do this, we used a mix of Tableau's built-in and Python-powered tools: linear regression with confidence bands, Pareto (80/20) analysis, k-means clustering, distribution bands, rolling averages, dual-axis charts, and several TabPy/Python calculated fields (z-score, statistical mode, variance, and Pearson correlation via NumPy). The dashboards are also interactive, with filter and highlight actions connecting the sheets.

All of this is built on the CMS HCRIS Hospital Cost Report data (2010–2024, 120,238 hospital-year records). While working with it, we found two data-quality issues that needed fixing before the numbers could be trusted: the raw file has duplicate CY/FY rows for almost every hospital-year (fixed by filtering to Reporting = CY), and a handful of records have clear data-entry errors in bed counts, staffing counts, and revenue (fixed with reasonable upper-bound filters, noted under each visualization below).

Operational Dashboard

Revenue vs. Bed Capacity — Regression Analysis

This scatter plot looks at whether bed capacity predicts net patient revenue, and whether that holds true across all hospital types or varies by type. Each point is one hospital in one fiscal year (58,841 records total, 2011–2023, CY-reporting only, capped at 5,000 beds to filter out data-entry errors). We fit a separate regression line for each of the three hospital types to compare them directly.

Key Regression Results

- Children's Hospitals see the biggest return — about \$2.84M in extra revenue per additional bed ($p < 0.0001$)
- Short-Term Acute Care hospitals gain roughly \$1.68M per bed ($p < 0.0001$)
- Critical Access Hospitals gain only about \$149K per bed ($p < 0.0001$) All three relationships are highly statistically significant — bed capacity is a genuine, reliable predictor of revenue in every hospital type. But the financial return on that capacity varies by a factor of nearly 19 between the highest and lowest type.

Patterns & Implications

This gap comes down to how differently these hospitals are reimbursed and what kind of care they provide. Children's hospitals handle specialized, high-intensity cases — NICU, pediatric oncology, complex surgery — so each bed brings in a lot more revenue. Critical Access hospitals, on the other hand, are small rural facilities that get reimbursed on a cost basis by design, specifically so they can stay viable even with low patient volume. That's why adding beds barely moves the needle for them financially.

Strategic Takeaways

- Capacity decisions should be made per facility type, not based on one system-wide revenue-per-bed number
- Pediatric capacity offers the clearest revenue upside per bed — Critical Access hospitals shouldn't be expected to fund expansion through volume growth alone
- As per the objectives, revenue-per-bed comparisons should be segmented by facility type to avoid misleading averages.

Pareto Analysis — Uncompensated Care Cost Concentration

This dual-axis Pareto chart ranks all 56 states/territories by total uncompensated-care (UC) cost, overlaying a cumulative-percentage line against a 20%-of-states marker and an 80%-of-cost reference line.

What the Chart Shows

- The first 20% of states by position (11.2 of 56) account for 63.58% of total uncompensated-care cost nationally.
- Reaching 80% of total UC cost requires 20 of the 56 states (35.7% of states) — the data is concentrated, but does not follow an exact 80/20 split.

Why This Matters

Uncompensated-care costs aren't spread evenly — a small group of states carries most of the sector's charity-care and bad-debt burden. This matters a lot for how DSH funding and state Medicaid policy should actually be targeted.

Strategic Takeaways

- DSH funding at both the federal and state level should prioritize the top-20 states identified here — that's where additional funding would do the most to offset uncompensated care.
- MedInsight's peer-benchmarking product should flag hospitals in states with high UC concentration for closer monitoring of margin sustainability.
- Because the distribution is concentrated but not a clean 80/20 split, policy responses should be state-specific rather than applying a single national threshold.

Hospital Performance Clusters — Occupancy vs. Staffing Intensity

We used Tableau's built-in k-means clustering to group hospitals based on two operational measures: bed occupancy rate and staffing ratio (employees per bed). This covers 59,201 hospital-year records (2011–2023, CY-only, with beds capped at 5,000, occupancy capped at 150%, and staffing ratio capped at 50 to filter out data-entry errors).

Cluster Profiles

- Cluster 1 (Lower Occupancy): occupancy generally below ~40%.
- Cluster 2 (Higher Occupancy): occupancy above ~40%.

Key Insight

Occupancy rate — not staffing intensity — is the dominant factor separating the two clusters. This suggests that within this dataset, how full a hospital runs is a more fundamental operational signature than how heavily staffed it is relative to its bed count; the two measures vary largely independently of one another.

Strategic Takeaways

- Capacity planning should treat occupancy and staffing intensity as two separate levers, not one combined efficiency score
- Lower-occupancy hospitals are the clearest candidates for volume growth or consolidation analysis

Executive Scorecard

Expense Anomaly Detection — TabPy Z-Score Analysis

This visualization flags hospitals whose year-over-year expense growth looks statistically abnormal compared to the national trend, using a Python calculation (SCRIPT_REAL, NumPy) run through Tableau's TabPy extension. Any hospital with an absolute z-score of 2 or higher is flagged as an outlier — meaning its growth is more than two standard deviations away from the national average for 2023.

Key Findings

- Heartland Regional Medical Center is the clear standout, with a z-score around 42.45 — far beyond anything else in the dataset
- Group Health Central Hospital follows at a distant second, around 22.65
- A handful of other hospitals — about 8 to 10 out of roughly 4,700 nationwide — also show abnormal expense growth, some unusually high and some unusually low

Strategic Takeaways

- The analytical model operates as an initial screening tool rather than a final determination; therefore, flagged hospitals require individual assessment before drawing conclusions regarding systemic cost trends.
- Utilizing a z-score baseline of 2 establishes a statistically rigorous and justifiable framework for MedInsight's continuous benchmark-alert application.
- This methodological approach—overlaying Python-driven anomaly detection on standard Tableau metrics—can be replicated for other key performance indicators, such as staffing, revenue, and margins, in subsequent benchmarking operations.

Distribution of Staffing Ratio by Hospital Type

This box-and-whisker plot, with mean and ± 1 standard deviation bands layered on top, shows how staffing ratio (employees per bed) is distributed across the three hospital types. It covers 58,619 hospital-year records (2011–2023), filtered to remove since the divide by zero would mess up the graph due to outliers.

Key Findings

- Short-Term Acute Care shows the widest spread, with the highest median staffing ratio and values extending up to ~40 employees per bed.
- Critical Access hospitals show a lower, tighter distribution, consistent with their smaller scale and simpler mix.
- Children's hospitals have the smallest sample and a comparatively tight distribution, with one notable high outlier (~36)

Strategic Takeaways

- Staffing benchmarks should always be applied within hospital type, never across types, given how different the typical ranges are.
- Hospitals falling outside the ± 1 SD band are reasonable candidates for a staffing-efficiency review.
- The wide Short Term Acute Care range suggests this category may benefit from a secondary segmentation (e.g., by bed-size tier) in future benchmarking work.

Financial Resilience — Margin Trend, Rolling Average & Revenue-Margin Correlation

This set of visualizations asks a direct question: has growing revenue actually led to better operating performance? A three-year rolling average separates the long-term trend from year-to-year noise, native and Python-computed statistics describe the shape of the margin distribution, and a Python-calculated Pearson correlation tests directly whether revenue scale and profit margin move together.

Key Findings

- Net patient revenue grew 68% between 2011 and 2022 (\$719.7B → \$1,211.4B), yet the weighted operating margin was negative in every calendar year of the series.
- The three-year rolling margin worsened steadily, from -2.17% (2015) to -4.87% (2022) — and continued falling in 2021 even though the single-year margin improved that year, because the severe 2020 result remained inside the rolling window.
- When we limit the data to CY-only records from 2011–2022, the picture is consistent: average margin of -3.18%, median of -2.98%. Those two numbers sitting so close together point to a structural problem across the sector, not a single bad year pulling the average down. The spread (1.35 percentage points standard deviation) is fairly tight too — and the one year that breaks from it, 2020 at -5.89%, is exactly the year you'd expect to be different, given the pandemic.
- Revenue and operating margin are strongly inversely correlated ($r \approx -0.68$ to -0.82 depending on year range, pending final confirmation once a reporting-consistency filter is finalized on the correlation worksheet) — years with the highest sector revenue were not the years with the strongest operating performance.

Strategic Takeaways

- Revenue growth alone shouldn't be used as a stand-in for financial health in MedInsight's client reports — margin trend and rolling-average stability tell a more accurate story.
- The consistent negative margin across nearly all years argues for structural cost-management recommendations, not just revenue-growth strategies, in client advisory engagements.
- 2020 should be treated as a distinct, exceptional shock year in any trend analysis — its inclusion or exclusion materially changes rolling-average interpretation for 2021–2022.

Key Challenges

- Reimbursement structure drives revenue inequality by facility type. The nearly 19-fold gap in revenue per bed between Children's and Critical Access hospitals shows a system that rewards specialized, high-acuity care far more than the rural safety-net role Critical Access hospitals are built for.
- Uncompensated care is geographically concentrated. A small group of states carries most of the sector's charity-care and bad-debt burden — national averages hide where the real strain actually is.
- Revenue growth has not resolved margin pressure. Sector-wide, the strong inverse correlation between revenue and margin — combined with a rolling average that has worsened even in years of improving annual results — indicates a cost-structure problem that revenue growth alone cannot fix.

Strategic Recommendations

- Break down all future peer comparisons and benchmarking by facility type — a single industry-wide average badly distorts both revenue potential and staffing norms.
- Direct policy focus and Disproportionate Share Hospital (DSH) funding specifically toward the top 20 states facing the highest uncompensated care costs, instead of allocating resources uniformly.
- Employ MedInsight's z-score anomaly detection as a permanent screening mechanism across all upcoming reporting cycles to systematically identify true cost outliers and data integrity concerns.
- Pivot strategic advice for clients away from revenue expansion and toward operating cost structures, as sector data indicates that revenue volume does not forecast—and can even negatively correlate with—operating margins.
- Keep using the same data-prep steps in future MedInsight studies — the CY-only filter and the outlier caps on occupancy, beds, and staffing ratio — since the raw HCRIS data has enough duplication and errors that skipping these would skew the results. Overall, the challenges facing U.S. hospitals aren't confined to any one facility type, state, or year — they're systemic. Addressing them means benchmarking, funding, and strategic decisions that account for that underlying structure, not just sector-wide averages.